236002: metadacite, Daisy Milano mine

(Kurnalpi Terrane, Eastern Goldfields Superterrane, Yilgarn Craton)

Location and sampling

Widgiemooltha (SH 51-14), Lake Lefroy (3235)
MGA Zone 51, 397568E 6566817N

Sampled on 15 January 2019

This sample was collected from the 64.35 – 65.12 m depth interval of drillcore 18EISD001, drilled in 2018 by Silver Lake Resources at the Daisy Milano mine (McGahren, 2018), with support from the Western Australian Government's Exploration Incentive Scheme (EIS). The drillhole is located on Mount Monger station, about 8.2 west of Mulga Dam, 4.2 km north of 28 Mile Dam, and 0.9 km east-northeast of Carowlyme Hill.

Geological context

The unit sampled is a metadacite (Fig. 1) of the Kurnalpi Terrane (Swager et al., 1990; Griffin, 1990; Groenewald et al., 2006). The Daisy Milano locality is on the southwestern flank of the north-northwest-trending, doubly plunging Bulong anticline, the core of which contains mineralized granitic intrusive rocks in a succession of clastic metasedimentary, mafic–ultramafic meta-igneous, and felsic to intermediate metavolcanic rocks, all of which are intruded by quartz–feldspar porphyry dykes and sills (McGahren, 2018). This sample was collected from a succession of apparently interbedded felsic metavolcanic rocks and metakomatiite, although most contacts are obscured by dolerite intrusions. A porphyritic metadacite, sampled from drillhole YDD-1 about 15.0 km to the west-northwest, yielded an igneous crystallization age of 2672 ± 12 Ma (GSWA 104940A, Nelson, 1995).

Petrographic description

The sample is an altered, fine- to medium-grained metavolcanic rock of dacitic or possibly andesitic composition (Fig. 1), consisting of about 80–85% groundmass, 15% plagioclase phenocrysts, <5% secondary sericite, and accessory magnetite, pyrite, chalcopyrite and zircon. The rock is strongly affected by a texturally destructive alteration assemblage of fine-grained polygonal quartz and coarser-grained masses of carbonate minerals, which has partially replaced the groundmass that had previously undergone milder alteration by chlorite. The sample exhibits an apparent coarse clastic texture, which is due to overprinting alteration rather than primary volcaniclastic processes. The 'clasts' commonly have sharp boundaries that are crosscut by plagioclase phenocrysts. The groundmass is feldspathic with a crude trachytic texture which has been mostly destroyed elsewhere by the quartz–carbonate alteration. Plagioclase phenocrysts are subhedral, equant to tabular, up to 2 mm long, and albitized. Magnetite grains are irregular, elongate, up to 0.2 mm in size, and partly replaced by non-reflective iron oxide minerals. Pyrite grains are subhedral, irregular, and up to 0.35 mm in size, and there are trace tiny chalcopyrite grains, both spatially associated with carbonate minerals. The rock is not deformed and metamorphic grade is very low.

Zircon morphology

Zircons isolated from this sample are colourless, anhedral to subhedral, and variably rounded. The crystals are up to 250 μm long, and equant to elongate, with aspect ratios up to 5:1. In cathodoluminescence (CL) images, concentric zoning is ubiquitous. A CL image of representative zircons is shown in Figure 2.

Analytical details

This sample was analysed on 19–20 September 2020, using SHRIMP-B. Thirteen analyses of the M257 standard were obtained during the session, of which 11 analyses indicated an external spot-to-spot (reproducibility) uncertainty of 1.17% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.41% (1σ). Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Twenty-two analyses were obtained from 22 zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 3).

Interpretation

The analyses are concordant to moderately discordant (Fig. 3), and form a single group, based on their $^{207}Pb^*/^{206}Pb^*$ ratios.

Group I comprises 22 analyses (Table 1), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2702 ± 6 Ma (MSWD = 1.1).

The date of 2702 ± 6 Ma for the 22 analyses in Group I is interpreted as the age of igneous crystallization.

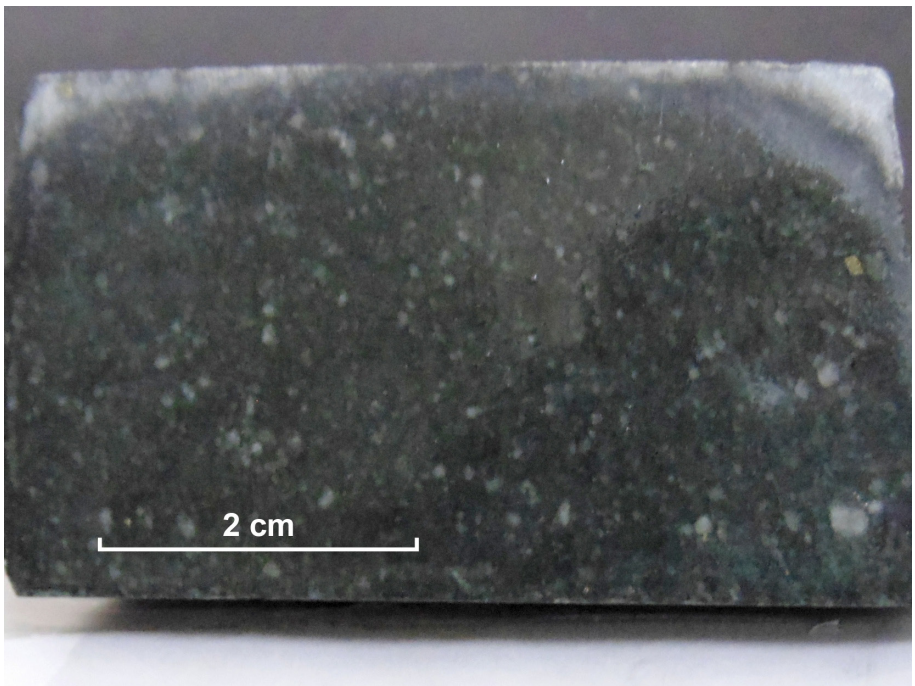

Figure 1. Drillcore image (wet surface) of sample 236002: metadacite, Daisy Milano mine

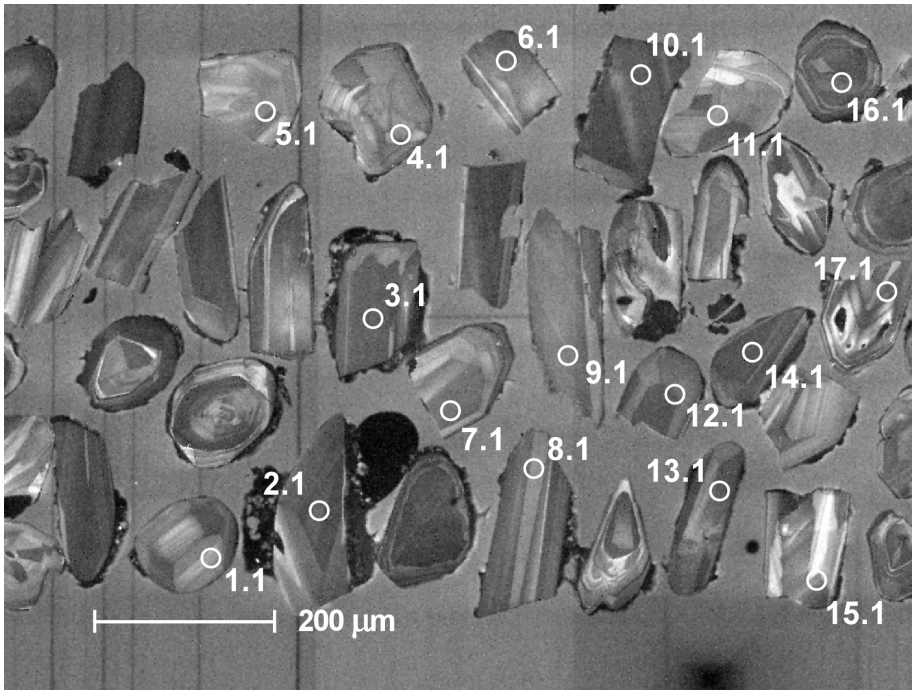

Figure 2. Cathodoluminescence image of representative zircons from sample 236002: metadacite, Daisy Milano mine. Numbered circles indicate the approximate locations of analysis sites

Table 1. Ion microprobe analytical results for zircons from sample 236002: metadacite, Daisy Milano mine

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb^*$ ± 1σ		$^{207}Pb^*/^{206}Pb^*$ ± 1σ		$^{238}U/^{206}Pb^*$ date (Ma) ±1σ		$^{207}Pb^*/^{206}Pb^*$ date (Ma) ±1σ		Disc. (%)
I	18	18.1	47	34	0.72	0.302	1.887	0.039	0.18478	0.00176	1.893	0.039	0.18209	0.00204	2735	47	2672	19	-2.3
I	5	5.1	43	27	0.63	0.378	1.911	0.039	0.18568	0.00174	1.918	0.039	0.18231	0.00208	2705	45	2674	19	-1.2
I	14	14.1	94	81	0.86	0.188	1.903	0.033	0.18403	0.00143	1.906	0.033	0.18235	0.00155	2718	39	2674	14	-1.6
I	21	21.1	127	76	0.60	0.106	1.984	0.032	0.18413	0.00179	1.986	0.032	0.18319	0.00183	2629	36	2682	17	2.0
I	13	13.1	67	53	0.79	0.098	1.951	0.036	0.18466	0.00154	1.953	0.036	0.18379	0.00162	2665	41	2687	15	0.8
I	7	7.1	65	51	0.78	0.156	1.899	0.035	0.18535	0.00150	1.902	0.035	0.18396	0.00163	2724	41	2689	15	-1.3
I	11	11.1	54	28	0.51	0.155	1.911	0.038	0.18544	0.00168	1.914	0.038	0.18405	0.00182	2710	44	2690	16	-0.7
I	12	12.1	79	57	0.73	0.153	1.900	0.034	0.18604	0.00137	1.903	0.034	0.18467	0.00148	2723	40	2695	13	-1.0
I	1	1.1	45	23	0.51	0.159	1.877	0.036	0.18618	0.00170	1.880	0.037	0.18476	0.00185	2749	44	2696	17	-2.0
I	9	9.1	65	65	1.01	0.140	2.047	0.038	0.18634	0.00160	2.050	0.038	0.18509	0.00172	2562	40	2699	15	5.1
I	20	20.1	134	102	0.77	0.050	1.900	0.031	0.18570	0.00110	1.901	0.031	0.18526	0.00114	2725	36	2701	10	-0.9
I	16	16.1	95	61	0.64	-0.042	1.911	0.033	0.18518	0.00123	1.910	0.033	0.18555	0.00126	2714	38	2703	11	-0.4
I	6	6.1	67	61	0.91	0.161	1.949	0.035	0.18706	0.00141	1.952	0.035	0.18562	0.00153	2666	40	2704	14	1.4
I	22	22.1	105	103	0.99	0.065	1.963	0.033	0.18648	0.00125	1.964	0.033	0.18590	0.00130	2653	37	2706	12	2.0
I	17	17.1	68	38	0.57	0.130	1.982	0.037	0.18713	0.00155	1.985	0.037	0.18597	0.00165	2630	41	2707	15	2.8
I	4	4.1	49	26	0.53	0.000	1.966	0.038	0.18629	0.00165	1.966	0.038	0.18629	0.00165	2651	43	2710	15	2.2
I	10	10.1	78	66	0.85	0.047	1.945	0.034	0.18687	0.00132	1.946	0.034	0.18645	0.00135	2674	39	2711	12	1.4
I	3	3.1	81	72	0.89	0.112	1.893	0.033	0.18767	0.00129	1.895	0.033	0.18667	0.00137	2732	39	2713	12	-0.7
I	2	2.1	78	73	0.94	0.069	1.904	0.032	0.18770	0.00131	1.905	0.032	0.18709	0.00136	2720	38	2717	12	-0.1
I	15	15.1	42	30	0.70	0.000	1.848	0.041	0.18719	0.00197	1.848	0.041	0.18719	0.00197	2788	50	2718	17	-2.6
I	19	19.1	77	76	0.99	0.074	1.941	0.035	0.18840	0.00135	1.942	0.035	0.18775	0.00140	2677	40	2723	12	1.7
I	8	8.1	51	32	0.63	0.078	1.879	0.037	0.18930	0.00171	1.880	0.037	0.18860	0.00178	2749	45	2730	16	-0.7

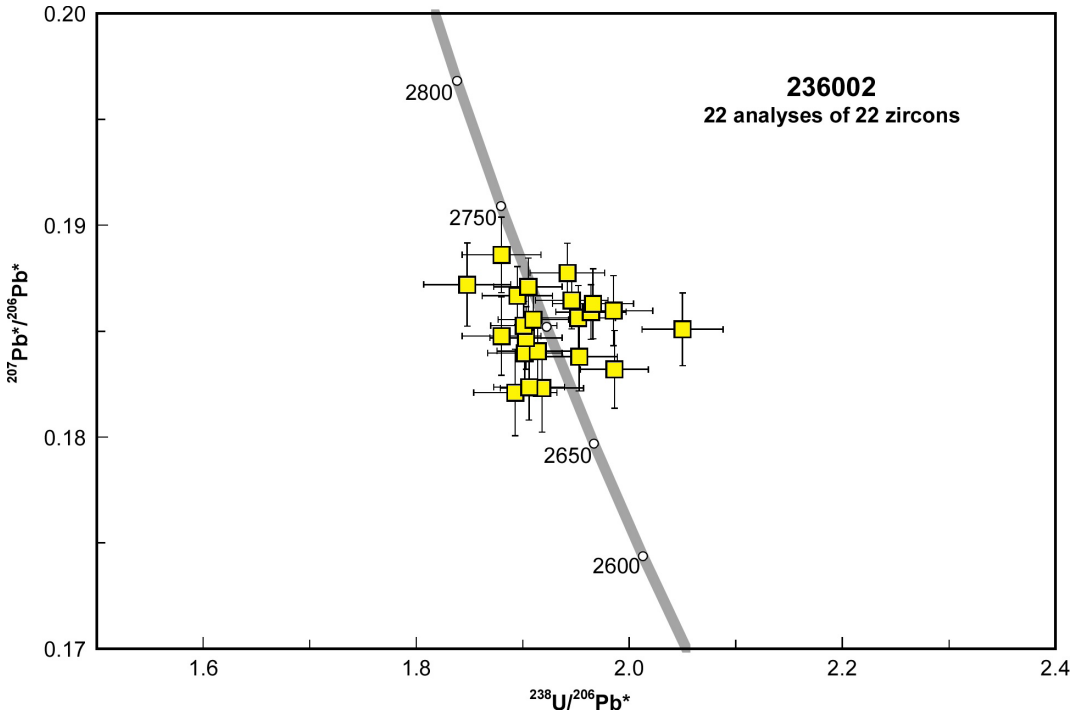

Figure 3. U–Pb analytical data for sample 236002: metadacite, Daisy Milano mine. Yellow squares indicate Group I (magmatic zircons)

References

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Recommended reference for this publication

Wingate, MTD, Lu, Y, Fielding, IOH and Smithies, RH 2022, 236002: metadacite, Daisy Milano mine; Geochronology Record 1857: Geological Survey of Western Australia, 4p.

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